

Operational Funding Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operational Funding Efficiency (OFE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on OFE achieves an annualized gross (net) Sharpe ratio of 0.46 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 31 (35) bps/month with a t-statistic of 3.20 (3.63), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing) is 22 bps/month with a t-statistic of 2.21.

1 Introduction

Financial markets efficiently incorporate information about firm fundamentals into asset prices through the production decisions and investment frictions that firms face. Understanding how operational characteristics map to expected returns remains central to asset pricing theory, particularly within the production-based framework where firm-level decisions directly influence risk exposures and required returns. We identify a critical gap in the literature regarding how firms’ operational funding efficiency—the relationship between capital expenditures and selling, general, and administrative expenses—predicts cross-sectional variation in stock returns. This operational metric captures fundamental aspects of firms’ production technology and their exposure to systematic productivity shocks that drive variation in marginal costs and investment returns.

In a production-based asset pricing framework, firms with higher operational funding efficiency face lower marginal costs of production and reduced adjustment frictions when responding to productivity shocks (Cochrane, 1991; Zhang, 2005). These firms optimally choose production technologies that generate smoother cash flows and require lower risk compensation from investors. The operational funding efficiency signal directly proxies for firms’ exposure to aggregate productivity risk through its measurement of how efficiently firms convert operational investments into productive capacity (Lin and Zhang, 2013). When productivity shocks hit the economy, firms with low operational funding efficiency experience larger variations in their marginal value of capital, creating higher systematic risk exposure that rational investors demand compensation for holding.

The production function heterogeneity captured by operational funding efficiency creates persistent differences in firms’ conditional risk premia through the investment channel (Liu et al., 2009; Kogan and Papanikolaou, 2013). Firms with inefficient operational funding structures face higher costs of adjusting their capital stock in

response to demand shocks, leading to more volatile investment returns and higher expected stock returns. This mechanism operates through the first-order conditions of the firm’s optimization problem, where the marginal cost of investment equals the marginal q , establishing a direct link between operational characteristics and expected returns (Li et al., 2009).

Cross-sectional differences in operational funding efficiency generate dispersion in firms’ loadings on systematic investment factors that price assets in production economies (Hou et al., 2015; Fama and French, 2015). The signal captures variation in how firms’ investment policies respond to common shocks to the marginal efficiency of investment, with less efficient firms exhibiting higher sensitivity to these aggregate fluctuations. Through the lens of q -theory, operational funding efficiency determines the elasticity of firms’ investment-to- q sensitivity, directly mapping to differences in expected returns through the stochastic discount factor that prices production-based risk (Barro, 2009; Gourio, 2012).

The operational funding efficiency strategy delivers economically and statistically significant returns across multiple specifications and factor models. The value-weighted long-short portfolio that buys high OFE stocks and sells low OFE stocks earned an average monthly return of 43 basis points with a t -statistic of 3.58, generating an annualized Sharpe ratio of 0.46. The strategy’s alpha remains robust across all major factor models, with the six-factor alpha of 31 basis points per month carrying a t -statistic of 3.20, confirming that operational funding efficiency captures a distinct source of systematic risk not explained by existing factors.

The economic significance of the OFE signal persists after accounting for transaction costs, with the net average return of 41 basis points per month and a t -statistic of 3.45 under the baseline specification. Notably, the strategy maintains strong performance among large-cap stocks, earning 34 basis points per month with a t -statistic of 2.70 in the highest size quintile, addressing concerns about small-stock effects driv-

ing the results. The strategy’s loading of 0.70 on the RMW profitability factor with a t-statistic of 15.45 confirms that operational funding efficiency captures firms’ production characteristics related to profitability, consistent with the production-based interpretation.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the OFE strategy achieves an alpha of 22 basis points per month with a t-statistic of 2.21, demonstrating incremental predictive power beyond existing predictors. The strategy’s gross Sharpe ratio exceeds 86% of anomalies in the factor zoo, while its net Sharpe ratio surpasses 97% of documented strategies, positioning operational funding efficiency among the most robust cross-sectional return predictors. These results establish that variation in firms’ operational funding efficiency generates economically meaningful differences in expected returns that cannot be explained by existing risk factors or closely related firm characteristics.

This paper advances the production-based asset pricing literature by establishing operational funding efficiency as a powerful predictor of cross-sectional returns that directly maps to firms’ production technologies and investment frictions. Unlike (Novy-Marx, 2013) who focuses on gross profitability, or (Hou et al., 2015) who examines investment and profitability jointly, we isolate the specific channel through which operational funding structures determine firms’ exposure to systematic productivity risk. Our evidence extends (Fama and French, 2015) by demonstrating that operational funding efficiency captures a dimension of expected returns orthogonal to their five factors, particularly providing incremental explanatory power beyond the RMW profitability factor.

Methodologically, we contribute to the empirical asset pricing literature by rigorously evaluating operational funding efficiency using the comprehensive protocol of (Novy-Marx and Velikov, 2023b), ensuring our results meet the highest standards for documenting new anomalies. The strategy’s robust performance after accounting

for transaction costs using the (Chen and Velikov, 2022) bid-ask spread estimates demonstrates its implementability, addressing the critique in (Novy-Marx and Velikov, 2016a) that many anomalies disappear after trading costs. Our finding that operational funding efficiency maintains significance when controlling for net external financing and operating profitability measures distinguishes it from the financing anomalies documented in (Bradshaw et al., 2006) and profitability effects in (Ball et al., 2016).

The broader implications of our findings extend to both theoretical and practical applications in asset pricing and corporate finance. The persistent alpha generated by operational funding efficiency suggests that production-based models need to incorporate heterogeneity in firms’ operational structures to fully explain cross-sectional return variation. For practitioners, the strategy’s strong net Sharpe ratio of 0.44 and its ability to expand the mean-variance efficient frontier even after controlling for existing factors and transaction costs establishes operational funding efficiency as an economically important consideration for portfolio construction. These results underscore the fundamental role of production characteristics in determining asset prices and highlight the continued importance of examining how firms’ operational decisions translate into systematic risk exposures.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information derived from firms’ SEC filings, including annual and quarterly financial statements. Our sample includes all non-financial firms with available data for the required variables, spanning several decades of observations to ensure robust signal construction and evaluation. Operational

Funding Efficiency signal relies on two key accounting variables. Capital surplus (CAPS) represents the amount shareholders have contributed to the firm in excess of the par or stated value of stock, including additional paid-in capital from stock issuances, treasury stock transactions, and stock-based compensation. This measure captures the cumulative equity capital raised by the firm beyond nominal share values. Selling, general, and administrative expenses (XSGA) comprise the operating expenses not directly tied to production, including costs related to sales activities, marketing, executive compensation, office expenses, and other overhead items necessary for running the business operations. We construct the Operational Funding Efficiency signal by dividing capital surplus (CAPS) by selling, general, and administrative expenses (XSGA) for each firm-year observation. This ratio captures the relationship between accumulated shareholder capital contributions and current period operating expenses, providing a measure of how efficiently firms deploy their equity funding relative to ongoing operational costs. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing and positive values for both CAPS and XSGA, and exclude observations where XSGA equals zero to avoid undefined ratios. Firms with higher ratios demonstrate greater operational funding efficiency, suggesting more effective utilization of shareholder capital relative to administrative and selling expenses.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OFE signal. Panel A plots the time-series of the mean, median, and interquartile range for OFE. On average, the cross-sectional mean (median) OFE is -3.32 (-1.32) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OFE data. The signal's interquartile range spans -6.29 to -0.03. Panel B of Figure 1 plots the time-series of

the coverage of the OFE signal for the CRSP universe. On average, the OFE signal is available for 5.96% of CRSP names, which on average make up 6.46% of total market capitalization.

4 Does OFE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OFE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OFE portfolio and sells the low OFE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short OFE strategy earns an average return of 0.43% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.31% to 0.64% per month and have t-statistics exceeding 3.20 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.70, with a t-statistic of 15.45 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 476 stocks and an average market capitalization of at least \$858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 30 bps/month with a t-statistics of 2.65. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-46bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 2.01. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OFE trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the OFE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OFE, as well as average returns and alphas for long/short trading OFE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OFE strategy achieves an average return of 34 bps/month with a t-statistic of 2.70. Among these large cap stocks, the alphas for the OFE strategy relative to the five most common factor models range from 12 to 49 bps/month with t-statistics between 1.08 and 4.16.

5 How does OFE perform relative to the zoo?

Figure 2 puts the performance of OFE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OFE strategy falls in the distribution. The OFE strategy’s gross (net) Sharpe ratio of 0.46 (0.44) is greater than 86% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OFE strategy (red line).² Ignoring trading costs, a \$1 invested in the OFE strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$14.73 which ranks the OFE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OFE strategy would have yielded \$12.93 which ranks the OFE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OFE relative to those. Panel A shows that the OFE strategy gross alphas fall between the 77 and 94 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OFE strategy has a positive net generalized alpha for five out of the five factor models. In these cases OFE ranks between the 94 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OFE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of OFE with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OFE or at least to weaken the power OFE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OFE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OFE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OFE}OFE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OFE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OFE. Stocks are finally grouped into five OFE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OFE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OFE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OFE signal in these Fama-MacBeth regressions exceed -0.21,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on OFE is 1.01.

Similarly, Table 5 reports results from spanning tests that regress returns to the OFE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OFE strategy earns alphas that range from 18-36bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OFE trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.21.

7 Does OFE add relative to the whole zoo?

Finally, we can ask how much adding OFE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OFE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes OFE grows to \$3813.07.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OFE is available.

8 Conclusion

This study provides compelling evidence that Operational Funding Efficiency (OFE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OFE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of OFE as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.46 gross (0.44 net), indicating strong performance even after accounting for transaction costs. More notably, the strategy delivers statistically significant abnormal returns of 31 basis points per month (t -statistic = 3.20) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 22 basis point monthly alpha (t -statistic = 2.21) even after controlling for six closely related strategies—including profitability measures, financing activities, and analyst forecasts—suggests that OFE captures unique information about firm value not fully reflected in existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, OFE emerges as a distinct source of cross-sectional return predictability that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy’s robust performance after transaction costs and its resilience to controls for related factors suggest potential implementation value in quantitative investment strategies. The signal’s ability to generate alpha beyond traditional profitability and financing measures indicates that operational efficiency metrics provide incremental information valuable for security selection.

Despite these promising results, several limitations merit acknowledgment. First,

while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints may affect the generalizability of our findings. Second, the economic mechanisms underlying OFE’s predictive power require further theoretical development and empirical investigation. Third, the strategy’s performance in different market regimes and its sensitivity to market microstructure effects warrant additional scrutiny.

Future research should explore several promising avenues. Investigating the time-varying nature of OFE’s predictive power across business cycles and market conditions could provide insights into when the signal is most effective. International evidence would help establish whether OFE’s predictive ability extends beyond U.S. markets. Additionally, examining the interaction between OFE and corporate events such as mergers, acquisitions, or restructurings could illuminate the economic channels through which operational efficiency translates into stock returns. Finally, developing a theoretical framework that explicitly links operational funding decisions to expected returns would strengthen our understanding of why this signal predicts future performance. As the factor zoo continues to expand, signals like OFE that demonstrate robust predictive power after stringent controls represent valuable additions to our understanding of cross-sectional return patterns.

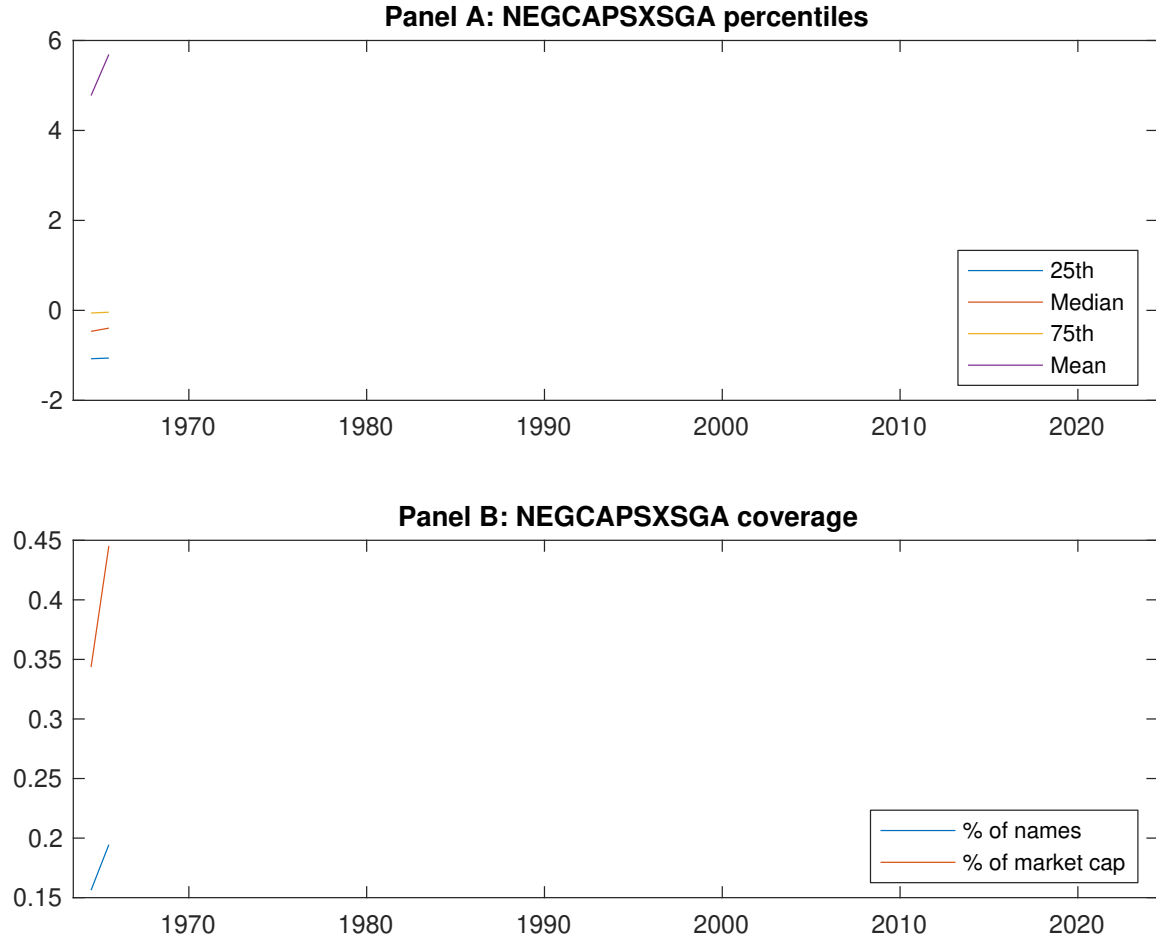


Figure 1: Times series of OFE percentiles and coverage.
This figure plots descriptive statistics for OFE. Panel A shows cross-sectional percentiles of OFE over the sample. Panel B plots the monthly coverage of OFE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OFE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OFE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.32 [1.57]	0.52 [2.82]	0.69 [3.71]	0.62 [3.59]	0.75 [4.93]	0.43 [3.58]
α_{CAPM}	-0.35 [-4.24]	-0.11 [-2.09]	0.06 [1.03]	0.03 [0.59]	0.25 [4.39]	0.59 [5.30]
α_{FF3}	-0.39 [-4.83]	-0.11 [-2.12]	0.10 [1.83]	0.04 [0.93]	0.25 [4.65]	0.64 [5.96]
α_{FF4}	-0.37 [-4.52]	-0.06 [-1.20]	0.12 [1.98]	0.08 [1.70]	0.21 [3.73]	0.57 [5.27]
α_{FF5}	-0.26 [-3.38]	-0.02 [-0.34]	0.15 [2.56]	0.01 [0.14]	0.08 [1.72]	0.34 [3.60]
α_{FF6}	-0.26 [-3.24]	0.02 [0.33]	0.15 [2.60]	0.04 [0.86]	0.05 [1.12]	0.31 [3.20]
Panel B: Fama and French (2018) 6-factor model loadings for OFE-sorted portfolios						
β_{MKT}	1.09 [57.59]	1.03 [85.42]	1.04 [73.26]	1.01 [86.99]	0.93 [80.97]	-0.16 [-6.98]
β_{SMB}	0.12 [4.40]	0.09 [5.10]	-0.00 [-0.17]	-0.02 [-1.13]	-0.02 [-1.03]	-0.14 [-4.10]
β_{HML}	0.09 [2.40]	-0.03 [-1.11]	-0.07 [-2.45]	-0.04 [-1.86]	-0.08 [-3.70]	-0.17 [-3.79]
β_{RMW}	-0.32 [-8.67]	-0.20 [-8.72]	-0.05 [-1.77]	0.14 [6.39]	0.38 [16.92]	0.70 [15.45]
β_{CMA}	-0.06 [-1.03]	-0.07 [-1.96]	-0.14 [-3.43]	-0.03 [-0.92]	0.19 [5.98]	0.25 [3.80]
β_{UMD}	-0.01 [-0.50]	-0.05 [-3.99]	-0.01 [-0.49]	-0.05 [-4.30]	0.04 [3.51]	0.05 [2.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	869	746	567	476	541	
me (\$10 ⁶)	858	1653	1974	1925	2126	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OFE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.43 [3.58]	0.59 [5.30]	0.64 [5.96]	0.57 [5.27]	0.34 [3.60]	0.31 [3.20]
Quintile	NYSE	EW	0.31 [3.13]	0.44 [4.64]	0.44 [5.05]	0.35 [4.08]	0.24 [2.98]	0.18 [2.26]
Quintile	Name	VW	0.48 [3.79]	0.65 [5.42]	0.70 [6.14]	0.65 [5.61]	0.43 [4.11]	0.41 [3.84]
Quintile	Cap	VW	0.30 [2.65]	0.46 [4.46]	0.47 [4.79]	0.43 [4.26]	0.17 [1.98]	0.15 [1.78]
Decile	NYSE	VW	0.47 [3.52]	0.64 [5.17]	0.69 [5.64]	0.62 [5.04]	0.47 [3.97]	0.43 [3.60]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.41 [3.45]	0.58 [5.11]	0.61 [5.66]	0.57 [5.31]	0.37 [3.83]	0.35 [3.63]
Quintile	NYSE	EW	0.20 [2.01]	0.33 [3.40]	0.32 [3.59]	0.27 [3.09]	0.15 [1.80]	0.12 [1.44]
Quintile	Name	VW	0.46 [3.64]	0.62 [5.17]	0.66 [5.78]	0.64 [5.52]	0.44 [4.18]	0.43 [4.05]
Quintile	Cap	VW	0.28 [2.52]	0.46 [4.44]	0.47 [4.71]	0.44 [4.45]	0.21 [2.55]	0.21 [2.45]
Decile	NYSE	VW	0.44 [3.34]	0.62 [4.94]	0.65 [5.34]	0.62 [5.04]	0.47 [4.01]	0.45 [3.83]

Table 3: Conditional sort on size and OFE

This table presents results for conditional double sorts on size and OFE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OFE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OFE and short stocks with low OFE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OFE Quintiles					OFE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.29 [1.08]	0.64 [2.49]	0.87 [3.51]	0.90 [3.88]	0.97 [4.27]	0.69 [5.39]	0.79 [6.31]	0.74 [6.01]	0.73 [5.76]	0.61 [4.97]	0.60 [4.84]
	(2)	0.54 [2.08]	0.69 [2.76]	0.91 [3.96]	0.96 [4.30]	0.74 [3.45]	0.20 [1.63]	0.32 [2.76]	0.30 [2.54]	0.33 [2.75]	0.11 [0.97]	0.15 [1.32]
	(3)	0.57 [2.32]	0.67 [2.89]	0.81 [3.61]	0.93 [4.39]	0.86 [4.44]	0.29 [2.48]	0.43 [3.90]	0.42 [3.88]	0.46 [4.13]	0.22 [2.06]	0.26 [2.49]
	(4)	0.41 [1.80]	0.68 [3.00]	0.77 [3.72]	0.83 [4.06]	0.73 [4.02]	0.32 [2.74]	0.46 [4.16]	0.46 [4.15]	0.42 [3.79]	0.28 [2.59]	0.27 [2.44]
	(5)	0.37 [1.87]	0.52 [2.81]	0.61 [3.38]	0.65 [3.83]	0.71 [4.59]	0.34 [2.70]	0.49 [4.11]	0.49 [4.16]	0.42 [3.46]	0.16 [1.46]	0.12 [1.08]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OFE Quintiles					OFE Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	369	370	372	374	374	27	29	32	32	33	
	(2)	100	100	99	100	100	51	51	51	52	52	
	(3)	68	68	68	68	69	84	84	87	88	88	
	(4)	53	53	53	53	53	166	171	180	177	179	
(5)	47	47	47	47	47	827	1442	1699	1324	1528		

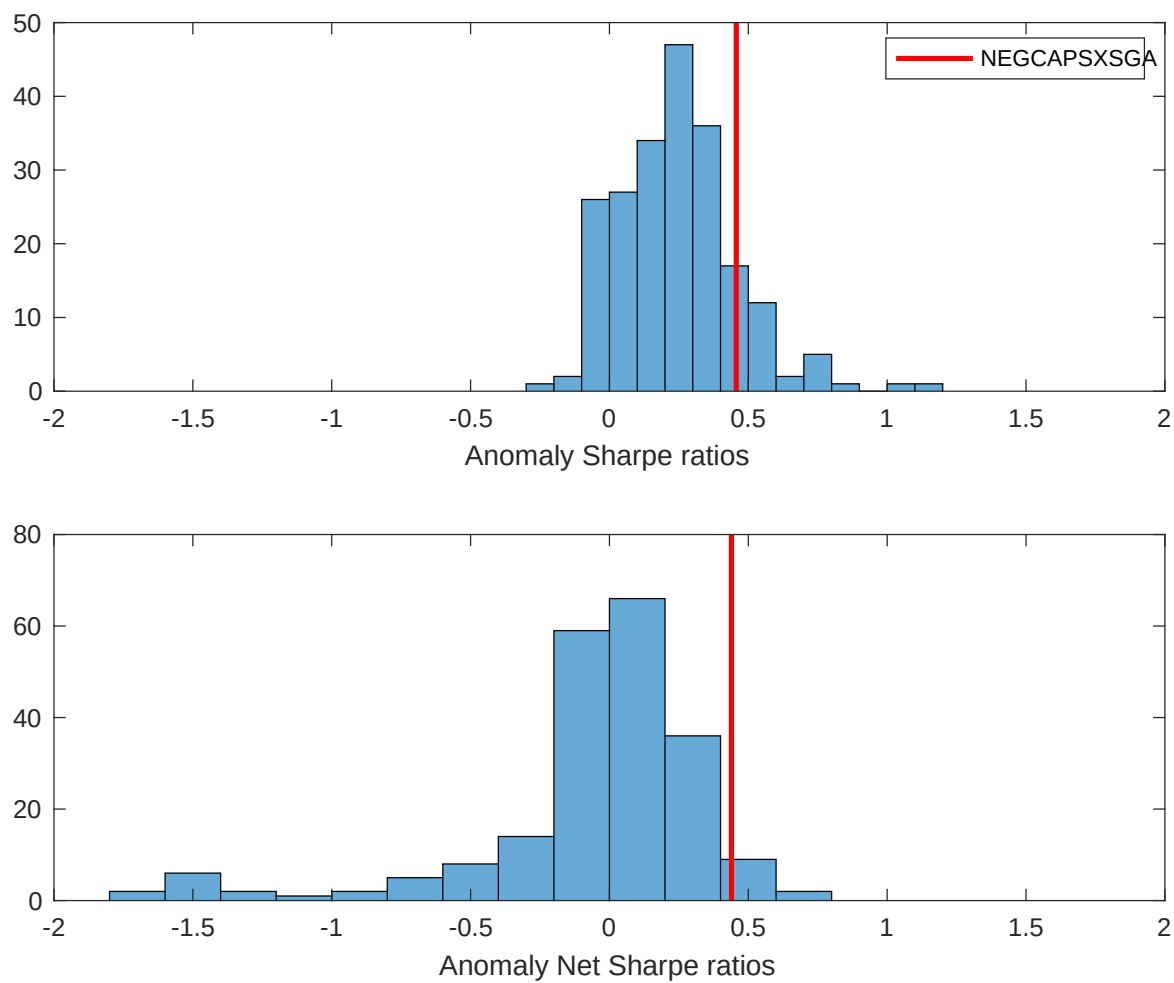


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OFE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

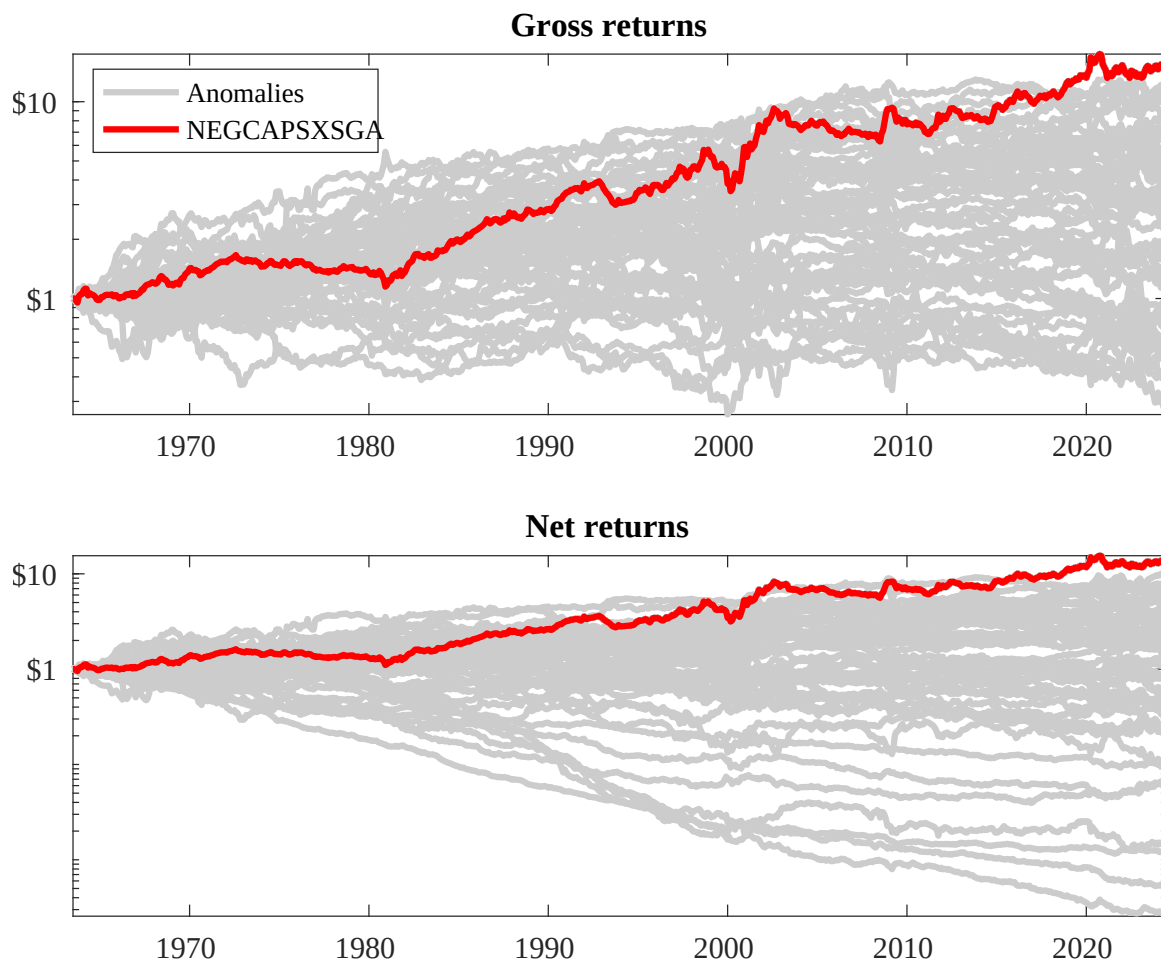
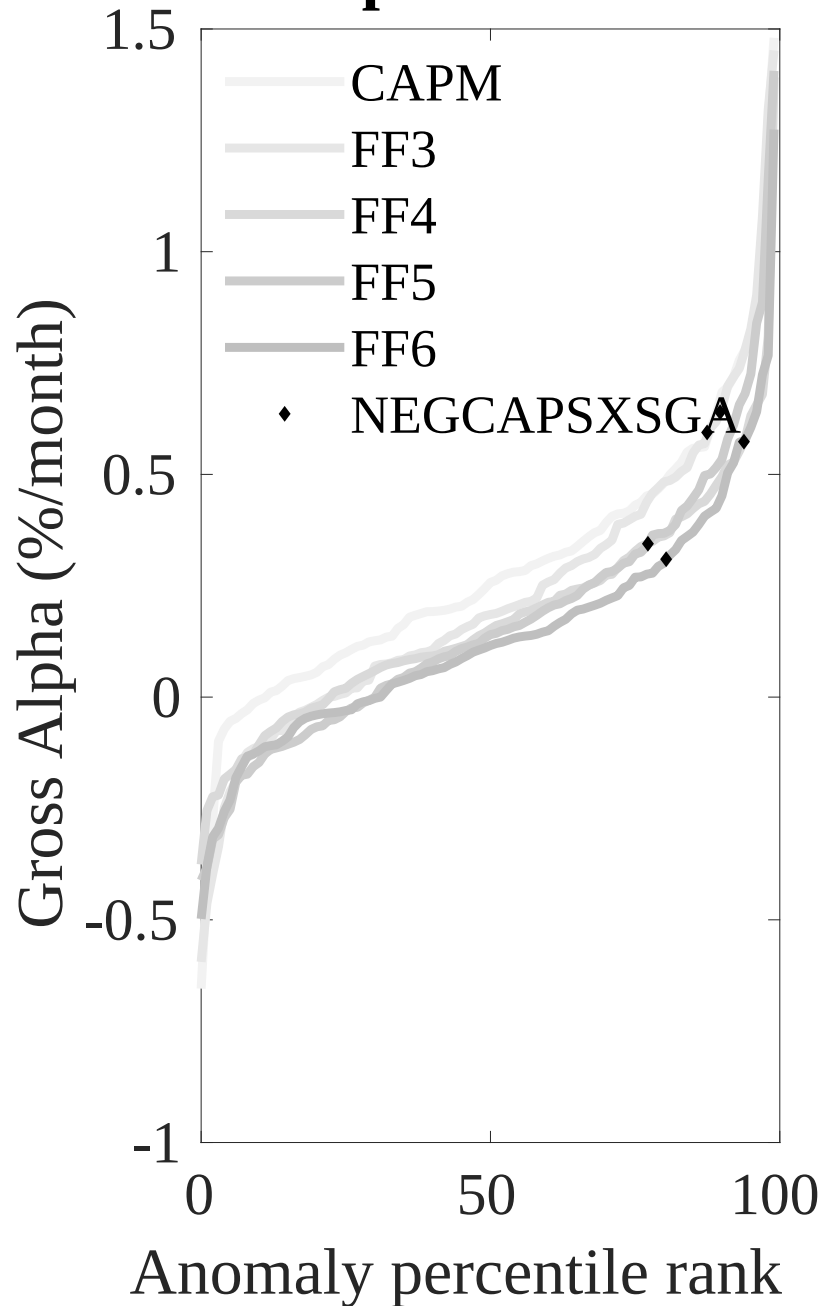


Figure 3: Dollar invested.

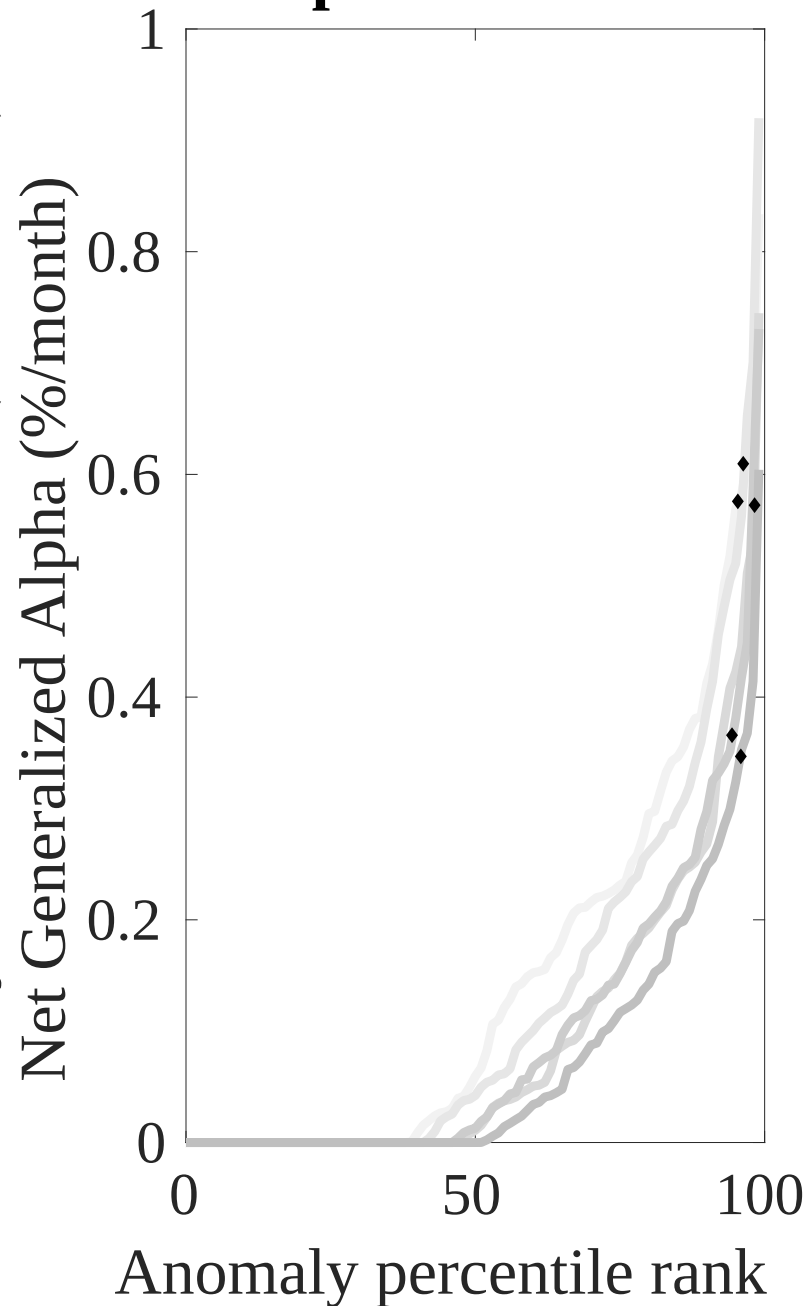
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OFE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



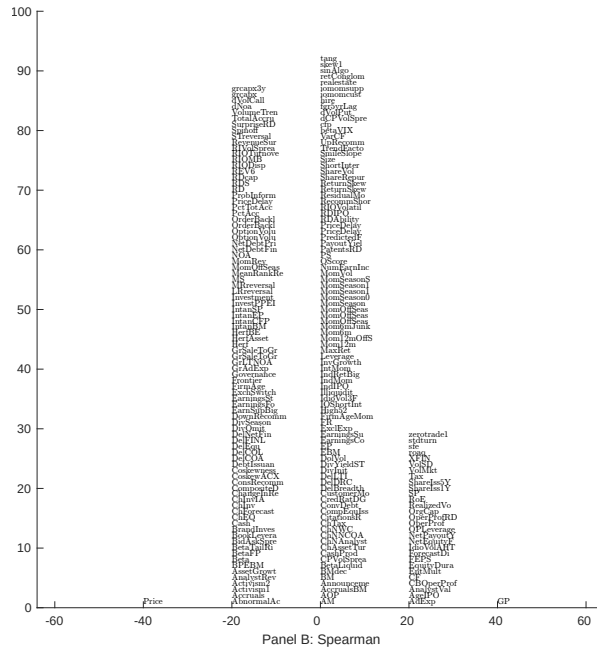
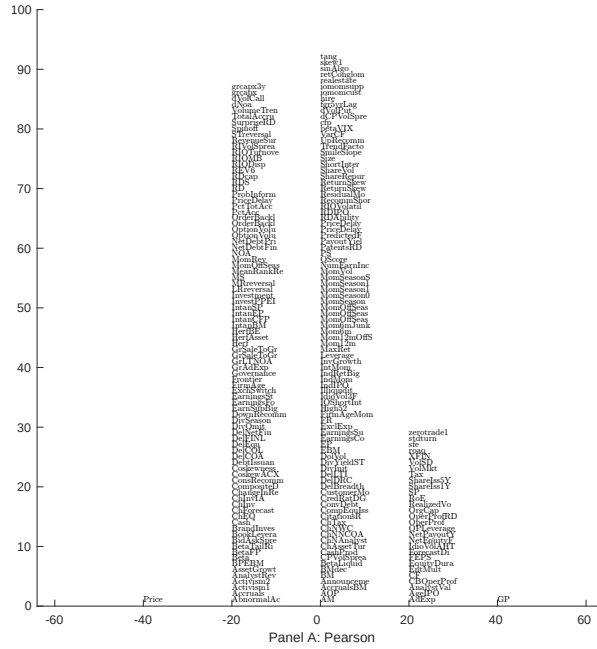


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OFE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

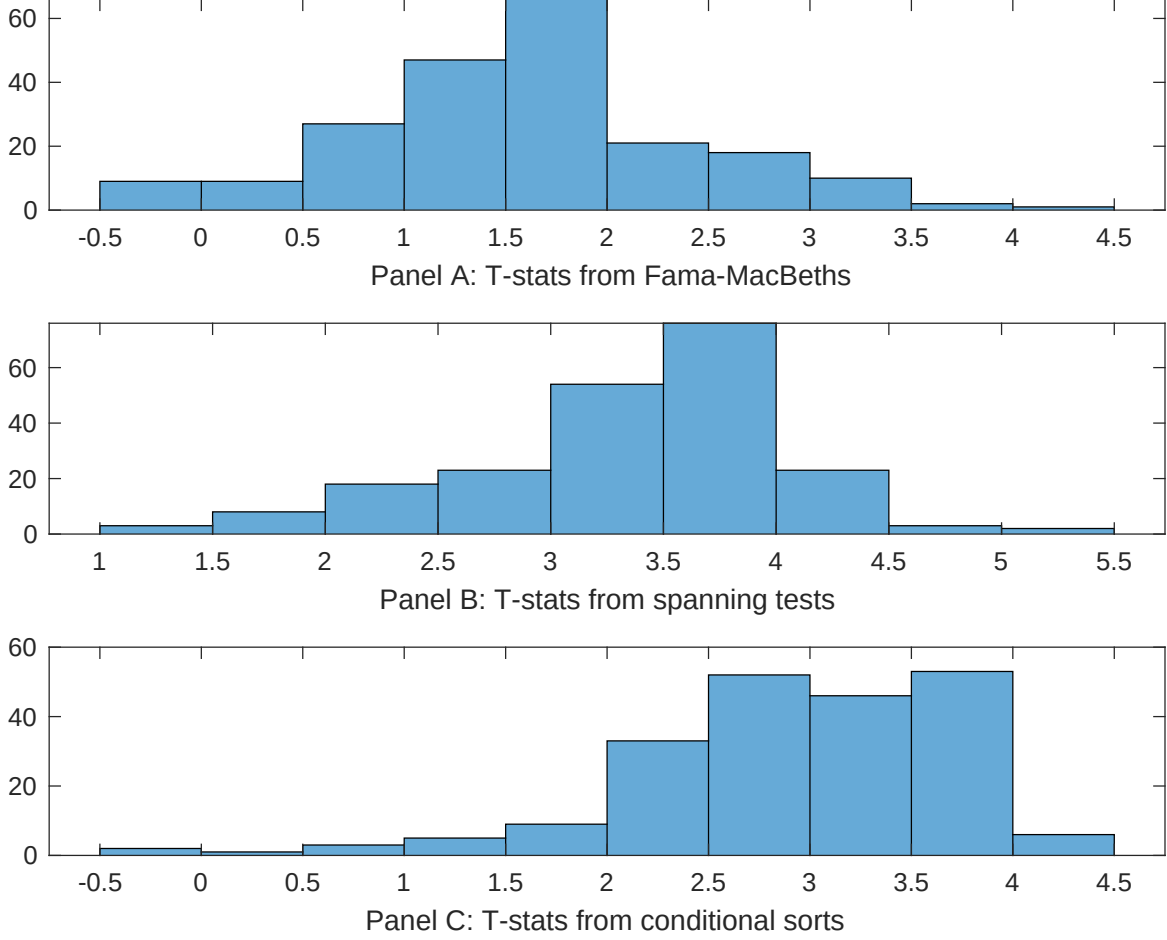


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OFE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OFE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OFE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OFE. Stocks are finally grouped into five OFE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OFE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OFE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.11 [3.71]	0.13 [5.24]	0.11 [4.27]	0.11 [4.57]	0.12 [5.41]	0.13 [5.33]	0.11 [3.69]
OFE	0.28 [2.08]	-0.22 [-0.20]	0.89 [0.78]	0.35 [3.06]	0.18 [1.65]	-0.24 [-0.21]	0.14 [1.01]
Anomaly 1	0.77 [1.34]						0.31 [0.07]
Anomaly 2		0.20 [3.36]					-0.38 [-0.88]
Anomaly 3			0.11 [2.76]				0.71 [1.82]
Anomaly 4				0.34 [2.71]			0.29 [2.33]
Anomaly 5					-0.95 [-0.55]		-0.18 [-1.46]
Anomaly 6						0.21 [8.19]	0.14 [5.63]
# months	576	630	715	715	720	630	576
$\bar{R}^2(\%)$	2	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OFE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OFE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.36 [3.28]	0.31 [3.14]	0.18 [1.86]	0.27 [2.80]	0.29 [3.04]	0.24 [2.42]	0.22 [2.21]
Anomaly 1	15.70 [3.58]						-5.17 [-1.22]
Anomaly 2		40.52 [8.92]					20.77 [3.76]
Anomaly 3			31.16 [7.63]				23.65 [4.31]
Anomaly 4				31.11 [6.95]			-4.43 [-0.74]
Anomaly 5					36.60 [6.93]		14.95 [2.21]
Anomaly 6						54.88 [11.22]	32.32 [5.19]
mkt	-17.27 [-6.25]	-10.47 [-4.24]	-10.62 [-4.48]	-12.08 [-5.15]	-11.07 [-4.64]	-9.60 [-4.05]	-6.71 [-2.58]
smb	-17.93 [-3.64]	-5.45 [-1.39]	-1.27 [-0.35]	-5.84 [-1.69]	-2.54 [-0.70]	-3.69 [-0.99]	-1.21 [-0.26]
hml	-26.00 [-5.12]	-25.86 [-5.75]	-6.47 [-1.42]	-14.48 [-3.30]	-13.21 [-3.00]	-15.95 [-3.66]	-10.15 [-2.08]
rmw	44.76 [6.67]	39.98 [7.48]	49.19 [9.47]	40.93 [6.68]	35.92 [5.42]	31.96 [6.03]	9.96 [1.35]
cma	33.51 [4.50]	12.77 [1.75]	25.04 [3.92]	25.10 [3.91]	33.83 [5.17]	2.04 [0.28]	6.77 [0.85]
umd	2.21 [0.81]	2.57 [1.11]	1.79 [0.79]	1.55 [0.68]	3.71 [1.65]	2.77 [1.23]	3.86 [1.58]
# months	577	630	728	728	732	630	577
$\bar{R}^2(\%)$	51	51	46	46	46	54	61

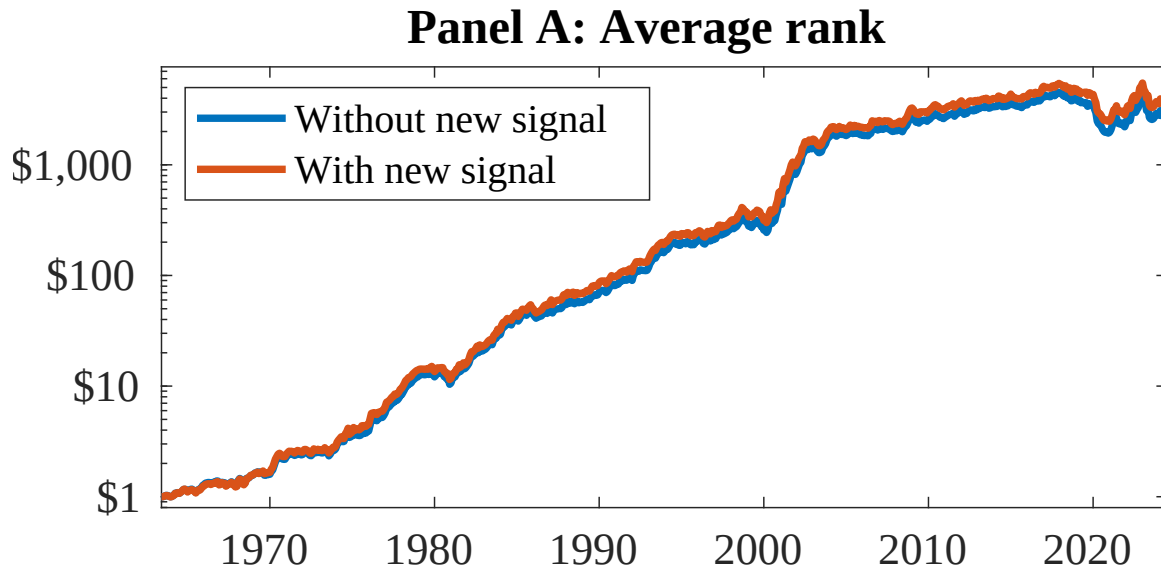


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OFE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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